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Work vehicle

Abstract

A work vehicle including: a transmission configured to receive motive power input from a motive power source, and shift and output the input motive power to a travel device; a transmission case housing the transmission; and a gear pair disposed in the transmission, in a bottom section of an internal space of the transmission case, and including (i) a first power transmission gear configured to rotate about a first rotation axis, and (ii) a second power transmission gear configured to mesh with the first power transmission gear and rotate about a second rotation axis that is parallel with the first rotation axis. The work vehicle further includes at least one gear cover covering the first power transmission gear and the second power transmission gear.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention The present invention relates to a work vehicle.

2. Description of the Related Art

(1) As shown in JP 2019-95058A, there is a work vehicle (a tractor) that includes: a transmission (a shift power transmission device, a forward and rearward travel switching device) configured to receive power input from a motive power source (an engine) and shift and output the input motive power to travel devices (front wheels, rear wheels); and a transmission case that houses the transmission.

SUMMARY OF THE INVENTION

(2) In the above-described work vehicle, in the case where the transmission is provided with a gear pair that includes a first power transmission gear that is rotatable about a first rotation axis and a second power transmission gear that meshes with the first power transmission gear and is rotatable about a second rotation axis that is parallel with the first rotation axis, and the gear pair is disposed in a bottom section of the internal space of the transmission case, the first power transmission gear and the second power transmission gear enter the lubricating oil stored in the transmission case and agitate the lubricating oil. Therefore, if the drive load applied to the first power transmission gear and the second power transmission gear due to the agitation of the lubricating oil increases, the

power transmission loss of the transmission increases.

(3) The present invention provides a work vehicle in which motive power can be efficiently transmitted through a transmission despite the first power transmission gear and the second power transmission gear agitating the lubricant oil.

(4) A work vehicle according to one aspect of the present invention includes: a transmission configured to receive motive power input from a motive power source, and shift and output the input motive power to a travel device; a transmission case housing the transmission; and at least one gear pair disposed in the transmission, in a bottom section of an internal space of the transmission case, and including (i) a first power transmission gear configured to rotate about a first rotation axis, and (ii) a second power transmission gear configured to mesh with the first power transmission gear and rotate about a second rotation axis that is parallel with the first rotation axis. The work vehicle further includes at least one gear cover covering the first power transmission gear and the second power transmission gear.

(5) With this configuration, the gear cover prevents the agitation of the lubricating oil by the first power transmission gear and the second power transmission gear from spreading into the lubricating oil storage area in the transmission case. Therefore, the drive load applied to the first power transmission gear and the second power transmission gear due to the agitation of the lubricating oil can be reduced, and the transmission can efficiently transmit motive power despite the first power transmission gear and the second power transmission gear agitating the lubricating oil.

(6) In the above aspect of the present invention, it is preferable that the at least one gear cover is configured to be splittable into a first split cover covering the first power transmission gear and a second split cover covering the second power transmission gear.

(7) With this configuration, when attaching the gear cover, it is possible to split the first split cover and the second split cover. Therefore, it is easier to attach the gear cover.

(8) In the above aspect of the present invention, it is preferable that the second split cover is integrated with a bottom wall of the transmission case.

(9) With this configuration, it is unnecessary to provide a special coupling structure for supporting the second split cover on the transmission case. Therefore, it is possible to support the second split cover on the transmission case at low cost.

(10) In the above aspect of the present invention, it is preferable that the bottom wall is configured to be detachable from a body of the transmission case.

(11) With this configuration, it is possible to attach the second split cover by attaching the bottom wall to the body of the transmission case. Therefore, it is easier to attach the second split case to the transmission case.

(12) In the above aspect of the present invention, it is preferable that the travel device includes a front wheel and a rear travel device, the transmission includes a front-wheel power transmission section configured to be switchable to (i) an equal-speed power transmission state, in which the front-wheel power transmission section transmits motive power to the front wheel in such a manner that a peripheral speed of the front wheel is equal to a peripheral speed of the rear travel device, and (ii) a speed-increasing power transmission state, in which the front-wheel power transmission unit transmits motive power to the front wheel in such a manner that the peripheral speed of the front wheel is higher than the peripheral speed of the rear travel device, the at least one gear pair includes a gear pair disposed in the front-wheel power transmission section and configured to realize the equal-speed power transmission state, and a gear pair disposed in the front-wheel power transmission section and configured to realize the speed-increasing power transmission state, and the at least one gear cover includes a gear cover covering the gear pair configured to realize the equal-speed power transmission state, and a gear cover covering the gear pair configured to realize the speed-increasing power transmission state.

(13) With this configuration, it is possible to enable the work vehicle to turn with a normal turning

radius by switching the front-wheel power transmission unit to the equal-speed power transmission state so that the peripheral speed of the front wheel is equal to the peripheral speed of the rear travel device, and enable the work vehicle to turn with a smaller turning radius than the normal turning radius by switching the front-wheel power transmission unit to the front-wheel speed-increasing power transmission state so that peripheral speed of the front wheel is higher than the peripheral speed of the rear travel device. Therefore, a turning radius can be selected based on the task, the work area, and so on. Despite the configuration that allows selection of turning radius, in the gear pair that realizes the equal-speed power transmission state and the gear pair that realizes the front-wheel speed-increasing power transmission state, the gear cover reduces the drive load applied to the first power transmission gear and the second power transmission gear due to the agitation of the lubricating oil, and accordingly enables the transmission to efficiently transmit motive power.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view showing the entirety of a tractor.
- (2) FIG. 2 is a schematic view showing a motive power transmission device.
- (3) FIG. 3 is a side view showing a front-wheel power transmission unit.
- (4) FIG. 4 is a cross-sectional view taken along the line IV-IV in FIG. 3.
- (5) FIG. 5 is a cross-sectional view taken along the line V-V in FIG. 3.
- (6) FIG. 6 is a side view showing a first gear cover, a second gear cover, and a clutch cover.
- (7) FIG. 7 is a front view showing the first gear cover.
- (8) FIG. 8 is a perspective view showing a first equal-speed split cover, the clutch cover, and a first speed-increasing split cover in a disassembled state.
- (9) FIG. 9 is a perspective view showing a second equal-speed split cover, a second speed-increasing split cover, and a bottom wall.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(10) Hereinafter, an embodiment of a tractor that is an example of a work vehicle according to the present invention will be described based on the drawings. In the embodiment, a front-rear direction with respect to a travel body of the tractor is defined as the direction in which the body travels in a working state thereof, and a left-right direction with respect to the body is defined as the left-right direction seen in the direction in which the body travels. Specifically, the direction indicated by the reference sign (F) in FIG. 1 is the forward direction with respect to the body, the direction indicated by the reference sign (B) in FIG. 1 is the rear ward direction with respect to the body, the direction indicated by the reference sign (U) in FIG. 1 is the upward direction with respect to the body, and the direction indicated by the reference sign (D) in FIG. 1 is the downward direction with respect to the body. The front side of the sheet of FIG. 1 corresponds to the left side of the body, and the back side of the sheet of FIG. 1 corresponds to the right side of the body. Therefore, the left-right direction with respect to the body corresponds to the width direction of the body.

(11) Entirety of Tractor

(12) FIG. 1 shows a tractor. This tractor includes a travel body 3 that is supported by a pair of left and right front wheels 1 (travel devices) that are steerable and drivable, and pair of left and right rear wheels 2 (travel devices) that are steerable. A prime mover 5 that includes an engine 4 is provided in a front section of the travel body 3. A driving section 6 on which an operator boards and operates the vehicle, and a linkage mechanism 7 that couples work devices such as a rotary tillage device with each other so that the work devices can be raised and lowered, are provided in a rear section of the travel body 3. The driving section 6 includes a driver's seat 8, a steering wheel 9

for steering the front wheels **1**, and a cabin **10** that covers the boarding space. Body frames **11** of the travel body **3** are constituted by the engine **4**, a transmission case **12** whose front section is coupled to a rear section of the engine **4**, front-wheel support frames **13** coupled to a lower section of the engine **4**, and so on. A PTO shaft **14** that transmits motive power from the engine **4** to the work devices coupled to each other by the linkage mechanism **7** is provided in a rear section of the transmission case **12**.

(13) Motive Power Transmission Device for Travel

(14) As shown in FIG. 2, a motive power transmission device **15** for travel, which transmits motive power to the front wheels **1** and the rear wheels **2**, includes a transmission **18** that shifts and transmits the motive power from the engine **4**, which serves as a motive power source, to a rear wheel differential mechanism **16** and a front wheel differential mechanism **17**. The transmission **18** is housed in the transmission case **12**.

(15) As shown in FIG. 3, the transmission **18** includes an input shaft **20** to which motive power from an output shaft **4a** of the engine **4** is transmitted via a main clutch **19**, a main transmission unit **21** to which motive power from the input shaft **20** is input, and that shifts and outputs the input motive power, a stepwise power transmission unit **22** to which the output from the main transmission unit **21** is input, a forward and rearward travel switching device **23** to which the output from the stepwise power transmission unit **22** is input, a gear interlocking mechanism **24** that transmits the output from the forward and rearward travel switching device **23** to an input shaft **16a** of the rear wheel differential mechanism **16**, and a front-wheel power transmission unit **25** to which the output from the forward and rearward travel switching device **23** is input.

(16) Main Transmission Unit

(17) As shown in FIG. 2, the main transmission unit **21** includes a continuously variable transmission device **28** to which motive power from the input shaft **20** is input via a rotation shaft **26** and a gear interlocking mechanism **27**, and a planetary transmission device **31** to which motive power from the input shaft **20** is input via a gear interlocking mechanism **29** and to which the output from the continuously variable transmission device **28** is input via a gear interlocking mechanism **30**.

(18) The continuously variable transmission device **28** includes a variable displacement hydraulic pump **P** to which motive power from the input shaft **20** is input via the rotation shaft **26** and the gear interlocking mechanism **27**, and a hydraulic motor **M** that is driven by pressure oil from the hydraulic pump **P**. By changing the swash plate angle of the hydraulic pump **P**, the continuously variable transmission device **28** shifts the motive power from the input shaft **20** into forward rotation motive power or reverse rotation motive power, steplessly changes the rotation speed of the forward rotation motive power or the reverse rotation motive power, and output the motive power from the hydraulic pump **P**. The continuously variable transmission device **28** is formed as a hydraulic static continuously variable transmission device.

(19) The planetary transmission device **31** includes a first planetary transmission unit **31a** to which the motive power from the input shaft **20** is input via the gear interlocking mechanism **29**, and to which the output from the continuously variable transmission device **28** is input via the gear interlocking mechanism **30**, and a second planetary transmission unit **31b** that is interlocked with the first planetary transmission unit **31a**. The interlocking of the first planetary transmission unit **31a** and the second planetary transmission unit **31b** is realized by the planetary gear of the first planetary transmission unit **31a** and the planetary gear of the second planetary transmission unit **31b** interlocking with each other. The planetary transmission device **31** is formed as a combined planetary transmission device.

(20) In the main transmission unit **21**, the continuously variable transmission device **28** shifts the motive power from the engine **4** into forward rotation motive power or reverse rotation motive power and steplessly changes the rotation speed of the forward rotation motive power or reverse rotation motive power, and the output from the continuously variable transmission device **28** and

the motive power from the engine **4** are combined with each other by the first planetary transmission unit **31a** and the second planetary transmission unit **31b** in the planetary transmission device **31**, and the combined motive power is output from three output shafts **32a**, **32b**, and **32c**.

(21) Stepwise Power Transmission Unit

(22) The stepwise power transmission unit **22** includes a first gear interlocking mechanism **33a** that is coupled to the first output shaft **32a** of the three output shaft **32a**, **32b**, and **32c** of the planetary transmission device **31**, a second gear interlocking mechanism **33b** that is coupled to the third output shaft **32c** of the three output shaft **32a**, **32b**, and **32c**, a third gear interlocking mechanism **33c** that is coupled to the second output shaft **32b** of the three output shaft **32a**, **32b**, and **32c**, and a fourth gear interlocking mechanism **33d** that is coupled to the third output shaft **32c** of the three output shaft **32a**, **32b**, and **32c**. The first gear interlocking mechanism **33a** is configured to be interlocked with and coupled to the output shaft **34** via a first clutch CL1. The second gear interlocking mechanism **33b** is configured to be interlocked with and coupled to the output shaft **34** via a second clutch CL2. The third gear interlocking mechanism **33c** is configured to be interlocked with and coupled to the output shaft **34** via a third clutch CL3. The fourth gear interlocking mechanism **33d** is configured to be interlocked with and coupled to the output shaft **34** via a fourth clutch CL4.

(23) In the stepwise power transmission unit **22**, when the first clutch CL1 is engaged, the continuously variable transmission device **28** is shifted and the steplessly shifted output from the planetary transmission device **31** is output from the output shaft **34** as motive power that has been shifted steplessly in a first speed range. When the second clutch CL2 is engaged, the continuously variable transmission device **28** is shifted and the steplessly shifted output from the planetary transmission device **31** is output from the output shaft **34** as motive power that has been shifted steplessly in a second speed range that is higher than the first speed range. When the third clutch CL3 is engaged, the continuously variable transmission device **28** is shifted and the steplessly shifted output from the planetary transmission device **31** is output from the output shaft **34** as motive power that has been shifted steplessly in a third speed range that is higher than the second speed range. When the fourth clutch CL4 is engaged, the continuously variable transmission device **28** is shifted and the steplessly shifted output from the planetary transmission device **31** is output from the output shaft **34** as motive power that has been shifted steplessly in a fourth speed range that is higher than the third speed range.

(24) Forward and Rearward Travel Switching Device

(25) The forward and rearward travel switching device **23** includes an input shaft **23a** that is coupled to the output shaft **34** of the stepwise power transmission unit **22**, and a forward travel clutch CLF and a rearward travel clutch CLR that are provided on the input shaft **23a**.

(26) The forward travel clutch CLF is configured to be interlocked with and coupled to an output shaft **23b** via a forward travel gear interlocking mechanism **35f**. The rearward travel clutch CLR is configured to be interlocked with and coupled to the output shaft **23b** via a rearward travel gear interlocking mechanism **35r**.

(27) In the forward and rearward travel switching device **23**, the output from the stepwise power transmission unit **22** is input to the input shaft **23a**, and when the forward travel clutch CLF is engaged, the motive power from the input shaft **23a** is converted by the forward travel gear interlocking mechanism **35f** to forward travel power, which is output from the output shaft **23b**, and when the rearward travel clutch CLR is engaged, the motive power from the input shaft **23a** is converted by the rearward travel gear interlocking mechanism **35r** to rearward travel power, which is output from the output shaft **23b**. The forward travel power and the rearward travel power output from the output shaft **23b** are input by the gear interlocking mechanism **24** to the rear wheel differential mechanism **16** and the front-wheel power transmission unit **25**.

(28) In the rear wheel differential mechanism **16**, the input motive power is transmitted from left and right output shafts **16b** to the left and right rear wheels **2**. The motive power from the left

output shaft **16b** is transmitted to the left rear wheel **2** via a planetary deceleration mechanism **36**. The left output shaft **16b** is provided with a steering brake **37**. Although not shown, the power transmission system from the right output shaft **16b** to the right rear wheel **2** is provided with the planetary deceleration mechanism **36** and the steering brake **37**, as with the power transmission system to the left rear wheel **2**.

(29) Front-Wheel Power Transmission Unit

(30) As shown in FIGS. **3** and **4**, the front-wheel power transmission unit is provided in a bottom section **12d** of the internal space of the transmission case **12**. As shown in FIG. **2**, the front-wheel power transmission unit **25** includes an input shaft **25a** that is coupled to an output shaft **24a** of the gear interlocking mechanism **24**, and an output shaft **25b** that is parallel with the input shaft **25a**. The input shaft **25a** and the output shaft **25b** are provided in such a manner as to extend in the front-rear direction of the transmission case **12** (the front-rear direction with respect to the body), and the output shaft **25b** is at a lower position than the input shaft **25a** with respect to the transmission case (with respect to the body of the tractor). The output shaft **24a** of the gear interlocking mechanism **24** is provided with a parking brake **38**.

(31) As shown in FIG. **2**, the input shaft **25a** is provided with an equal-speed clutch CLT and a speed-increasing clutch CLH that is rearward of the equal-speed clutch CLT with respect to the transmission case.

(32) An equal-speed gear pair **39** is provided over the equal-speed clutch CLT and the output shaft **25b**. The equal-speed gear pair **39** includes a first equal-speed power transmission gear **39a** and a second equal-speed power transmission gear **39b**, where the first equal-speed power transmission gear **39a** is coupled to an output-side member of the equal-speed clutch CLT and is rotatable about a first rotation axis X1, which is the axis of the input shaft **25a**, and the second equal-speed power transmission gear **39b** is provided on the output shaft **25b**, meshes with the first equal-speed power transmission gear **39a**, and is rotatable about a second rotation axis X2, which is the axis of the output shaft **25b**. The equal-speed gear pair **39** is configured to realize an equal-speed power transmission state in which the peripheral speed of the front wheels **1** is equal to the peripheral speed of the rear wheels **2**, by controlling the power transmission gear ratio between the first equal-speed power transmission gear **39a** and the second equal-speed power transmission gear **39b**.

(33) A speed-increasing gear pair **40** is provided over the speed-increasing clutch CLH and the output shaft **25b**. The speed-increasing gear pair **40** includes a first speed-increasing power transmission gear **40a** and a second speed-increasing power transmission gear **40b**, where the first speed-increasing power transmission gear **40a** is coupled to an output-side member of the speed-increasing clutch CLH and is rotatable about the first rotation axis X1, which is the axis of the input shaft **25a**, and the second speed-increasing power transmission gear **40b** is provided on the output shaft **25b**, meshes with the first speed-increasing power transmission gear **40a**, and is rotatable about the second rotation axis X2, which is the axis of the output shaft **25b**. The speed-increasing gear pair **40** is configured to realize a front-wheel speed-increasing power transmission state in which the peripheral speed of the front wheels **1** is higher than the peripheral speed of the rear wheels **2** by controlling the power transmission gear ratio between the first speed-increasing power transmission gear **40a** and the second speed-increasing power transmission gear **40b**.

(34) As shown in FIGS. **3**, **4**, and **5**, the first equal-speed power transmission gear **39a**, the second equal-speed power transmission gear **39b**, the first speed-increasing power transmission gear **40a**, and the second speed-increasing power transmission gear **40b** are provided in the bottom section **12d** of the internal space of the transmission case **12**. The first equal-speed power transmission gear **39a** and the second equal-speed power transmission gear **39b** are covered by a first gear cover **41**, and the first speed-increasing power transmission gear **40a** and the second speed-increasing power transmission gear **40b** are covered by a second gear cover **42**.

(35) In the front-wheel power transmission unit **25**, when the equal-speed clutch CLT is engaged, the motive power from the input shaft **25a** is transmitted by the equal-speed clutch CLT and the

equal-speed gear pair **39** to the output shaft **25b**, the equal-speed gear pair **39** realizes an equal-speed power transmission state, and motive power that drives the front wheels **1** is output from the output shaft **25b** so that the peripheral speed of the front wheels **1** is equal to the peripheral speed of the rear wheels **2**. When the speed-increasing clutch CLH is engaged, the motive power from the input shaft **25a** is transmitted by the speed-increasing clutch CLH and the speed-increasing gear pair **40** to the output shaft **25b**, the speed-increasing gear pair **40** realizes a front-wheel speed-increasing power transmission state, and motive power that drives the front wheels **1** is output from the output shaft **25b** so that the peripheral speed of the front wheels **1** is higher than the peripheral speed of the rear wheels **2**. The output from the output shaft **25b** is input to the front wheel differential mechanism **17** via a rotation shaft **43** that couples the output shaft **25b** and an input shaft **17a** of the front wheel differential mechanism **17**.

(36) In the travel body **3**, when the equal-speed clutch CLT is engaged, a four-wheel drive state in which the front wheels **1** and the rear wheels **2** are driven so that the average peripheral speed of the left and right front wheels **1** is equal to the average peripheral speed of the left and right rear wheels **2** is realized, and when the speed-increasing clutch CLH is engaged, a four-wheel drive state in which the front wheels **1** and the rear wheels **2** are driven so that the average peripheral speed of the left and right front wheels **1** is higher than the average peripheral speed of the left and right rear wheels **2** is realized. As a result, when the speed-increasing clutch CLH is engaged, the travel body **3** is enabled to turn with a smaller turning radius than when the equal-speed clutch CLT is engaged.

(37) When the equal-speed clutch CLT is engaged, the first equal-speed power transmission gear **39a** and the second equal-speed power transmission gear **39b** are rotated by the motive power from the input shaft **25a**, and the first speed-increasing power transmission gear **40a** and the second speed-increasing power transmission gear **40b** are rotated by the motive power from the output shaft **25b**. When the speed-increasing clutch CLH is engaged, the first speed-increasing power transmission gear **40a** and the second speed-increasing power transmission gear **40b** are rotated by the motive power from the input shaft **25a**, and the first equal-speed power transmission gear **39a** and the second equal-speed power transmission gear **39b** are rotated by the motive power from the output shaft **25b**. Although the lubricating oil stored in the transmission case **12** is agitated by the first equal-speed power transmission gear **39a**, the second equal-speed power transmission gear **39b**, the first speed-increasing power transmission gear **40a**, and the second speed-increasing power transmission gear **40b**, the first gear cover **41** prevents the agitation of the lubricating oil by the first equal-speed power transmission gear **39a** and the second equal-speed power transmission gear **39b** from spreading into the lubricating oil storage area. The second gear cover **42** prevents the agitation of the lubricating oil by the first speed-increasing power transmission gear **40a** and the second speed-increasing power transmission gear **40b** from spreading into the lubricating oil storage area.

(38) First Gear Cover and Second Gear Cover

(39) As shown in FIGS. **3**, **4**, **6**, and **7**, the first gear cover **41** is split into a first equal-speed split cover **41A** that covers the first equal-speed power transmission gear **39a** and a second equal-speed split cover **41B** that covers the second equal-speed power transmission gear **39b**. As shown in FIGS. **3**, **5**, and **6**, the second gear cover **42** is split into a first speed-increasing split cover **42A** that covers the first speed-increasing power transmission gear **40a** and a second speed-increasing split cover **42B** that covers the second speed-increasing power transmission gear **40b**.

(40) As shown in FIGS. **3**, **4**, **6**, **7**, and **8**, the first equal-speed split cover **41A** includes a front cover section **44** that covers the first equal-speed power transmission gear **39a** from the front side, and a circumference cover section **45** that covers the first equal-speed power transmission gear **39a** from the outer circumference thereof. The outer peripheral edge of the front cover portion **44** and the front end of the circumference cover section **45** are coupled to each other. The front cover section **44** and the circumference cover section **45** are coupled through welding. A lower section of the

circumference cover section **45** is provided with an opening **46** from which a lower section of the first equal-speed power transmission gear **39a** protrudes toward the second equal-speed power transmission gear **39b**.

(41) As shown in FIGS. **3**, **5**, **6**, and **8**, the first speed-increasing split cover **42A** includes a rear cover section **48** that covers the first speed-increasing power transmission gear **40a** from the rear side, and a circumference cover section **49** that covers the first speed-increasing power transmission gear **40a** from the outer circumference thereof. The circumference cover section **49** is integrated with a rear end of a clutch cover **50** that covers the equal-speed clutch CLT and the speed-increasing clutch CLH from the outer circumferences thereof. The rear end of the circumference cover section **49** and the outer peripheral edge of the rear cover section **48** are coupled to each other. The coupling of the circumference cover section **49** and the rear cover section **48** to each other is realized by coupling front-rear coupling bolts **51** (see FIGS. **6** and **8**), which is provided for the circumference cover section **49**, to the rear cover section **48**. A lower section of the circumference cover section **49** is provided with an opening **52** from which a lower section of the first speed-increasing power transmission gear **40a** protrudes toward the second speed-increasing power transmission gear **40b**.

(42) As shown in FIGS. **3** and **6**, an attachment hole **44a** (see FIG. **8**), which is provided in the front cover section **44** of the first equal-speed split cover **41A**, is fitted onto a bearing **53** that supports a front section of the input shaft **25a**, and the first equal-speed split cover **41A** is supported by the bearing **53**. An attachment cylinder section **54** (see FIG. **8**), which is provided in the rear cover section **48** of the first speed-increasing split cover **42A**, is fitted onto a bearing **55** that supports a rear section of the input shaft **25a**, and the first speed-increasing split cover **42A** is supported by the bearing **55**. As shown in FIG. **3**, a coupling bolt **56** couples a coupling section **50a** (see FIG. **8**), which is provided on an upper section of the clutch cover **50**, to a supporting section **12a**, which is provided in the transmission case **12**. As shown in FIG. **6**, a coupling bolt **47** (see FIG. **8**) couples the first equal-speed split cover **41A** and the clutch cover **50** to each other, and the clutch cover **50** prevents the first equal-speed split cover **41A** from being rotated and positions the first equal-speed split cover **41A** with respect to the first equal-speed power transmission gear **39a** in the direction along the first rotation axis **X1**. The first speed-increasing split cover **42A** and the clutch cover **50** are coupled to each other, and the clutch cover **50** prevents the first speed-increasing split cover **42A** from being rotated, and positions the first speed-increasing split cover **42A** with respect to the first speed-increasing power transmission gear **40a** in the direction along the first rotation axis **X1**.

(43) As shown in FIGS. **3**, **4**, **6**, **7**, and **9**, the second equal-speed split cover **41B** includes a front cover section **57** that covers the second equal-speed power transmission gear **39b** from the front side, a rear cover section **58** that covers the second equal-speed power transmission gear **39b** from the rear side, and a circumference cover section **59** that covers the second equal-speed power transmission gear **39b** from the outer circumference thereof. The front cover section **57** is provided with a through hole **57a** into which the output shaft **25b** is inserted, and the rear cover section **58** is provided with a cutout **58a** through which the output shaft **25b** is inserted. An upper section of the circumference cover section **59** is provided with an opening **60** from which an upper section of the second equal-speed power transmission gear **39b** protrudes toward the first equal-speed power transmission gear **39a**.

(44) As shown in FIGS. **3**, **4**, **5**, **6**, and **9**, the second speed-increasing split cover **42B** includes a front cover section **61** that covers the second speed-increasing power transmission gear **40b** from the front side, a rear cover section **62** that covers the second speed-increasing power transmission gear **40b** from the rear side, and a circumference cover section **63** that covers the second speed-increasing power transmission gear **40b** from the outer circumference thereof. The front cover section **61** is provided with a cutout **61a** into which the output shaft **25b** is inserted, and the rear cover section **62** is provided with a through hole **62a** through which the output shaft **25b** is

inserted. An upper section of the circumference cover section **63** is provided with an opening **64** from which an upper section of the second speed-increasing power transmission gear **40b** protrudes toward the first speed-increasing power transmission gear **40a**.

(45) As shown in FIG. **3**, **6**, **9**, and so on, the second equal-speed split cover **41B** and the second speed-increasing split cover **42B** are supported by a bottom wall **65** of the transmission case **12**. The second equal-speed split cover **41B** and the second speed-increasing split cover **42B** are supported by the bottom wall **65** in such a manner that the second equal-speed split cover **41B** and the second speed-increasing split cover **42B** are integrated with the bottom wall **65**. As shown in FIGS. **4**, **5**, **6**, and **9**, the bottom wall **65** is at a position between the rear cover section **58** of the second equal-speed split cover **41B** and the front cover section **61** of the second speed-increasing split cover **42B**, and includes a coupling section **65c** that couples the rear cover section **58** and the front cover section **61** to each other, and a horizontal section **65d** that is connected to an upper section of the coupling section **65c** and has a flange **66**. The coupling section **65c** has an internal space through which the output shaft **25b** passes. The width of the coupling section **65c** is smaller than the outer diameter of the second equal-speed split cover **41B** and the second speed-increasing split cover **42B**.

(46) The bottom wall **65** is configured to be detachable from a clutch housing **12c** of a body **12b** of the transmission case **12**. The clutch housing **12c** houses the equal-speed clutch CLT and the speed-increasing clutch CLH. Specifically, as shown in FIGS. **5**, **6**, **9**, and so on, the bottom wall **65** has the flange **66** provided on an upper section of the bottom wall **65**, and coupling bolts **67** (see FIG. **7**), which are inserted from below into coupling holes **66a** (see FIG. **9**) provided at multiple positions in the flange **66**, engage with the clutch housing **12c**, and accordingly the bottom wall **65** is coupled to the clutch housing **12c** of the body **12b** by the coupling bolts **67**.

(47) As shown in FIGS. **3** and **6**, the bottom wall **65** includes a front supporting section **65a** into which a front section of the output shaft **25b** is inserted via a bearing **68**, and a rear supporting section **65b** into which a rear section of the output shaft **25b** is inserted via a bearing **68**, and the bottom wall **65** is configured so that the front supporting section **65a** and the rear supporting section **65b** support the output shaft **25b**. The output shaft **25b** is supported by the bottom wall **65**. With this configuration, the second speed-increasing split cover **42B** positions the second speed-increasing power transmission gear **40b** and the second equal-speed power transmission gear **39b** with respect to a rear section of the bottom wall **65**, and the second equal-speed split cover **41B** positions the second speed-increasing power transmission gear **40b** and the second equal-speed power transmission gear **39b** with respect to a front section of the bottom wall **65**. In addition, the second equal-speed split cover **41B** is positioned with respect to the second equal-speed power transmission gear **39b** in the direction along the second rotation axis **X2**, and the second speed-increasing split cover **42B** is positioned with respect to the second speed-increasing power transmission gear **40b** in the direction along the second rotation axis **X2**. It is possible to attach the second equal-speed power transmission gear **39b**, the second speed-increasing power transmission gear **40b**, the second equal-speed split cover **41B**, and the second speed-increasing split cover **42B** to the transmission case **12** by attaching the bottom wall **65** to the body **12b**.

(48) (1) Although the above embodiment describes an example in which the front wheels **1** and the rear wheels **2** are provided, a crawler travel device or a mini crawler may be employed as a travel device.

(49) (2) Although the above embodiment describes an example in which the equal-speed gear pair **39** and the speed-increasing gear pair **40** are provided as a gear pair, the present invention is not limited to such a configuration, and a gear pair that transmits motive power only to the rear wheels **2** or a gear pair that simply transmits motive power without changing the transmission speed thereof may be employed.

(50) (3) Although the above embodiment describes an example in which the first gear cover **41** can be split into the first equal-speed split cover **41A** and the second equal-speed split cover **41B** and

the second gear cover **42** can be split into the first speed-increasing split cover **42A** and the second speed-increasing split cover **42B**, the present invention is not limited to such a configuration, and unsplitable cover may be employed.

(51) (4) Although the above embodiment describes an example in which the second equal-speed split cover **41B** and the second speed-increasing split cover **42B** are integrated with the bottom wall **65**, split covers that are separate from the bottom wall **65** and are not supported by the bottom wall **65**, or split covers that are separate from the bottom wall **65** and are supported by the bottom wall **65** may be employed.

(52) (5) Although the above embodiment describes an example in which the bottom wall **65** is detachable from the clutch housing **12c**, a bottom wall that is integrated with the clutch housing **12c** and cannot be detached from the clutch housing **12c** may be employed.

(53) (6) Although the above embodiment describes an example in which the motive power source is the engine **4**, an electric motor may be employed as a power source.

(54) (7) The present invention is applicable to a transmission that outputs power to a travel device, and a work vehicle provided with a transmission case that houses the transmission.

DESCRIPTION OF REFERENCE SIGNS

(55) **1** Front Wheel (Travel Device) **2** Rear Wheel (Rear Travel Device) **4** Engine (Motive Power Source) **12** Transmission Case **12b** Body **12d** Bottom Section **18** Transmission **25** Front-Wheel Power Transmission Unit **39** Equal-Speed Gear Pair (Gear Pair) **39a** First Equal-Speed Power Transmission Gear (First Power Transmission Gear) **39b** Second Equal-Speed Power Transmission Gear (Second Power Transmission Gear) **40** Speed-Increasing Gear Pair (Gear Pair) **40a** First Speed-Increasing Power Transmission Gear (First Power Transmission Gear) **40b** Second Speed-Increasing Power Transmission Gear (Second Power Transmission Gear) **41** First Gear Cover (Gear Cover) **41A** First Equal-Speed Split Cover (First Split Cover) **41B** Second Equal-Speed Split Cover (Second Split Cover) **42** Second Gear Cover (Gear Cover) **42A** First Speed-Increasing Split Cover (First Split Cover) **42B** Second Speed-Increasing Split Cover (Second Split Cover) **65** Bottom Wall

Claims

1. A work vehicle comprising: a transmission configured to receive motive power input from a motive power source, and shift and output the input motive power to a travel device; a transmission case housing the transmission; and at least one gear pair disposed in the transmission, in a bottom section of an internal space of the transmission case, and including (i) a first power transmission gear configured to rotate about a first rotation axis, and (ii) a second power transmission gear configured to mesh with the first power transmission gear and rotate about a second rotation axis parallel with the first rotation axis, wherein the work vehicle further comprises at least one gear cover covering the first power transmission gear and the second power transmission gear, and wherein the at least one gear cover includes a first split cover covering at least an entire upper portion of the first power transmission gear in a circumferential direction of the first power transmission gear, and a second split cover covering at least a lower portion of the second power transmission gear in a circumferential direction of the second power transmission gear.
2. The work vehicle according to claim 1, wherein the second split cover is provided as a portion of a bottom wall of the transmission case.
3. The work vehicle according to claim 2, wherein the bottom wall is configured to be detachable from a body of the transmission case.
4. The work vehicle according to claim 1, wherein the first split cover further includes a front cover section and a rear cover section, wherein a clutch cover provided in between the front cover section

and the rear cover section, and wherein the front cover section, the rear cover section, and the clutch cover extend along a rotation axis of the first power transmission gear.
